

Practice Quiz: Systems — Everything Is Connected

Part A: Multiple Choice

1. A **system** is best described as: A) A single thing working on its own B) A collection of connected parts that affect each other C) A set of rules you have to follow D) Another word for a computer program

Answer: B — A system has parts, connections, and a purpose that the parts can't achieve alone.

1. What are the three things every system has? A) A start, a middle, and an end B) Parts, connections, and a purpose C) Inputs, outputs, and errors D) People, places, and things

Answer: B — Parts, connections between those parts, and a purpose are what define a system.

1. Your alarm doesn't go off, so you skip breakfast, can't focus in class, and bomb a quiz. This is an example of: A) Bad luck B) A ripple effect in a system C) A broken boundary D) A random event

Answer: B — One change in the morning routine system rippled through and affected everything else.

1. A **boundary** in a system is: A) A wall that nothing can cross B) The line between what's inside the system and what's outside C) The end of the system's life D) A rule that can never be broken

Answer: B — Boundaries separate inside from outside, though they can be fuzzy like a cell membrane.

1. Which of the following is NOT an example of a system? A) A basketball team running a play B) Your family during dinner C) A single grain of sand sitting alone D) Your school's daily schedule

Answer: C — A single isolated object with no connections isn't a system. Systems require connected parts.

1. Your best friend eats dinner at your house three times a week. This shows that system boundaries can be: A) Impossible to define B) Fuzzy — letting some things in and keeping others out C) Completely rigid D) Invisible to everyone

Answer: B — Boundaries aren't always walls; they can be more like cell membranes.

1. When your school decides to start 30 minutes later and bus schedules, sports, and homework all change, this is an example of: A) A system crash B) Boundaries disappearing C) A ripple effect through the system D) The system breaking permanently

Answer: C — One change rippled through the entire school system affecting many connected parts.

1. Why is a basketball team more than just five individual players? A) They wear matching uniforms B) They share a court C) They communicate, coordinate, and create something together none could do alone D) They are all the same height

Answer: C — The team is a system because the players connect and achieve something no single player could.

1. Understanding systems helps you: A) Avoid all problems forever B) Predict consequences before you act C) Control other people D) Eliminate all surprises

Answer: B — Seeing systems gives you the ability to think about what might ripple before you act.

1. Which is the best example of a ripple effect in your social media life? A) You post a photo and nothing happens B) You post a comment, someone screenshots it, it gets shared, and now everyone is talking about it C) You scroll through your feed quietly D) You turn off your phone

Answer: B — One action (posting a comment) rippled through the social media system.

Part B: Short Answer

1. Pick a system you are part of (friend group, sports team, family) and identify its parts, connections, and purpose. How would it change if one part was removed?
2. Think about your morning routine as a system. Describe two things that could go wrong and trace how each one ripples through the rest of your morning.
3. Describe a time when someone's action in your friend group or class created a ripple effect that surprised you. What system was involved?
4. How are the boundaries of your friend group different from the boundaries of your family? Give a specific example of something that crosses one boundary but not the other.
5. If your school added a new rule (like no phones during lunch), trace at least three ripple effects that change could cause through the school system.